



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India

(Autonomous College Affiliated to University of Mumbai)

Re- Examination Synoptic

July 2019

Max. Marks: 100

Class: SE

Course Code: IT44

Name of the Course: Operating Systems

Duration: 180 Minutes

Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks	CO
Q 1 (a)	Explain 7 different functions of operating system. Each function 1 mark. Or Differentiate between system call and function call . Diagram 2 marks. Each difference 1 mark.	7	1
Q 1 (b)	Explain in detail monolithic kernel architecture in detail with neat diagram.	5	1
Q. 1(c)	Explain RAID 0 and RAID 1 RAID explanation 1 marks. Diagram 1 mark.	4	2
Q.1 (d)	Given the following queue -- 95, 180, 34, 119, 11, 123, 62, 64 with the Read-write head initially at the track 50 and the tail track being at 199.Solve using FCFS. Calculate seek length and average seek length. Show neat diagram. Ans :640 tracks.-1/2 mark, average seek length 1/2 mark. Diagram 1 mark. Calculation 2 mark.	4	2
Q2 (a)	What is PCB? Why it is needed. Explain with neat diagram. What is PCB- 1 mark, its need 2 mark, diagram 1 mark. PCB explanation 6 mark OR Name different types of scheduler? Explain their role? Scheduler name 3 marks. Diagram 1 mark. Each scheduler explanation 2 mark.	10	2



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Q2 (b)	Solve the following using FCFS and RR scheduling with time quantum 4 ms. Calculate turn around time, waiting time, average turn around time, average waiting time, completion time ? <table><tr><td>Process</td><td>Arrival time</td><td>Burst time</td></tr><tr><td>P1</td><td>0</td><td>8</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>9</td></tr><tr><td>P4</td><td>3</td><td>5</td></tr></table> Gantt chart 1 mark Average waiting and average turn around time each 1/2 mark. Completion time each process 1/4 mark. Waiting time each process 1/4 mark Turn around time each process 1/4 mark.	Process	Arrival time	Burst time	P1	0	8	P2	1	4	P3	2	9	P4	3	5	10	2
Process	Arrival time	Burst time																
P1	0	8																
P2	1	4																
P3	2	9																
P4	3	5																
Q3 (a)	Explain bankers algorithm with suitable example. What is need of bankers algorithm 1 mark. Algorithm 5 mark. Example 4 mark.	10	3															
Q3 (b)	A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution? Justify your answer. Ans: X and Y be the processes. X increments x by 1 and Z decrements x by 2. Now, Z stores back and after this X stores back. So, final value of x is 1 and not -1 and two Signal operations make the semaphore value 2 again. So, now W and Z can also execute like this and the value of x can be 2 which is the maximum possible in any order of execution of the processes. Answer 2 marks, justification 3 marks. Or A solution to the Dining Philosophers Problem which avoids deadlock is ----- (a) ensure that all philosophers pick up the left fork before the right fork (b) ensure that all philosophers pick up the right fork before the left fork	5	3															

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	(c) ensure that one particular philosopher picks up the left fork before the right fork, and that all other philosophers pick up the right fork before the left fork (d) None of the above . Ans c. Yes, Dining-Philosophers imposes the restriction that a philosopher may pick up chopsticks only if both of them are available. It is correct restriction. Answer 2 marks, justification 3 marks.		
Q 3 (c)	What is deadlock. Explain necessary and sufficient conditions for deadlock. Deadlock 1 mark. 4 condition 4 mark.	5	3
Q4 (a)	Explain thrashing with neat diagram. Consider the following reference string 0,2,1,6,4,0,1,0,3,1,2,1. Apply the FIFO page replacement policy .Calculate miss and hit ratio for frame size 4. •Fault Rate = 9 / 12 = 0.75	10	4
Q4 (b)	Explain the working of TLB with neat diagram. Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____. $T(\text{eff}) = \text{hit ratio} * (\text{TLB access time} + \text{Main memory access time}) + (1 - \text{hit ratio}) * (\text{TLB access time} + 2 * \text{main memory time})$ $= 0.6 * (10 + 80) + (1 - 0.6) * (10 + 2 * 80)$ $= 0.6 * (90) + 0.4 * (170)$ $= 122.$ OR A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The	10	4



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	minimum size of the TLB tag is: Size of a page = 4KB = 2^{12} Total number of bits needed to address a page frame = $32 - 12 = 20$ If there are 'n' cache lines in a set, the cache placement is called n-way set associative. Since TLB is 4 way set associative and can hold total 128 (2^7) page table entries, number of sets in cache = $2^{7/4} = 2^5$. So 5 bits are needed to address a set, and 15 ($20 - 5$) bits are needed for tag. Formula 1 mark. Final answer 1 mark Steps 3 mark.		
Q5 (a)	Explain virtual memory with neat diagram. Ans: Need of virtual memory 2 mark. Diagram 2 mark. Explanation 6 marks.	5	4
Q 5 (c)	Explain file system software architecture. Diagram 1 mark. Architecture explanation 4 mark.	5	5
Q5 (b)	Explain unix file system. Diagram 2 mark. File system explanation 8 marks. OR Explain record blocking in detail. What is record blocking 1 mark, Its type 1 mark Diagram 3 marks. Explanation 5 mark	10	5